SCSI1113 Computational Mathematics

~~TUTORIAL 3~~ Assignment 5

~~Semester 2 Session 2015/2016~~ Sem 1 20212022

Due Date:

1. Calculate error value, ε and relative error, ε_{rel} of the following number and state either the approximate value is a good approximation or not.
   a) N = 2.71828182, n = 2.7182
   b) N = 98.350, n = 98.000
   c) N = 0.000068, n = 0.00006
   d) N = $2.317^{10^{-4}}$, n = 0.00023

2. Find the most accurate value of the following operation
   a) 2.2525 - 1.112 + 0.9856 + 3.3335
   b) 0.0111 + 0.02 + 0.003 + 0.0035
   c) 50.36 - 41.13 + 0.001 - 13.393
   d) 0.9999 - 0.9999 - 0.001 + 0.0001
   e) 2.212 - 1.1 + 2.231

   where each number has been rounded.

3. Give exact value *x* = 71.67659 and approximate value $\bar{x}$ = 71.7. Define and calculate
   a) error
   b) absolute error
   c) relative error

4. Given the following functions and their interval.
   - $3x - x^{-1} - 0$ for interval [0.25 dan 0.27]
   - $x^2 + 4x^2 - 10 - 0$ for interval [1, 2]

   Find the root for each function at 3 decimal points.
   a) Bisection method
   b) False method

5. Given a function, $f(x) = x^2 - 10x + 23$
   a) Using graph, approximate interval [*a,b*] so that *f(a)* and *f(b)* are in opposite direction. Make sure *a* and *b* are integer numbers.
   b) Based on the interval [*a,b*] obtained from (a), find value *x* where *f(x)* = 0, at 2 decimal points using bisection method and false method.

6. Given non linear function $f(x) = x^2 - 2x - 1$, find the root at 4 decimal points using:
   a) Newton method with $x_0 = 2.6$,
   b) Secant method with $x_0 = 2.6$ and $x_1 = 2.5$

7. Find root for the function $(x) = (x-2)^2 - \ln(x)$, using Secant method. Approximate values are at $x_0 = 0.5$ and $x_1 = 2.0$. Perform calculation at five decimal points.

8. Given a non linear function, $f(x) = 2x^3 - 0.5x + 5$
   Using Newton method, find root at 4 decimal points with $x_0 = -2$